Laplace–Resolvent Semi-Analytics for a Ratio Barrier Payoff under Kou Dynamics

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1 Introduction

Decentralized finance (DeFi) lending protocols such as AAVE and Compound have opened a new class of capital-efficient trading strategies: a user deposits a cryptocurrency as collateral, borrows a second asset against it, and simultaneously sells the borrowed asset short. The result is a leveraged long–short position whose performance depends on the relative price dynamics of the two assets. With combined total value locked across major DeFi platforms exceeding $10 billion, these strategies have become a meaningful component of on-chain market activity, yet their risk profile remains analytically under-studied.

The defining feature of DeFi lending is automatic liquidation. Each platform computes a real-time *health factor* — a ratio of the collateral value to the outstanding liability — and liquidates the position the instant this ratio falls below a contractual threshold. Unlike traditional margin accounts, where a broker may issue a call and allow time for top-up, smart-contract liquidation is instantaneous and irrevocable. A practitioner must therefore choose the initial portfolio weights carefully: a more aggressive position amplifies expected returns on surviving paths but raises the probability of forced exit. The central question this paper addresses is how to set those weights optimally given the underlying asset dynamics.

Wrong-way risk is the dominant concern. An adverse move in either leg — a fall in the collateral asset or a rise in the borrowed asset — simultaneously erodes the numerator and inflates the denominator of the health factor. Cryptocurrency markets are particularly exposed: daily log-returns of 10–20% are common and flash-crash episodes can exceed 50%, so fat-tailed jump dynamics are essential for capturing realistic liquidation probabilities.

We model the bivariate log-price process under the *Kou double-exponential jump–diffusion* framework (Kou, 2002), in which each asset follows a diffusion with correlated Brownian parts and independent compound Poisson jumps with double-exponentially distributed sizes. This specification captures the asymmetric, leptokurtic return distributions observed in cryptocurrency markets while remaining analytically tractable. In particular, exponential jump sizes are eigenfunctions of the generator, which allows the Laplace-domain resolvent equation to be solved in closed form.

Measure change. Computing the killed moments $\mathbb{E}[\Pi_T^k \mathbf{1}_{\{\tau>T\}}]$ directly would require tracking the joint distribution of the bivariate process (X_1, X_2) up to the killing time, a substantially harder object. A change of measure eliminates this complexity. Multiplying the path measure by $e^{kX_{2,T}}/\mathbb{E}[e^{kX_{2,T}}]$ absorbs the k-th power of S_2 into the new numeraire, so that under the k-tilted measure $\mathbb{P}^{(k)}$ the payoff factor $e^{kX_{2,T}}$ disappears from the expectation and the remaining integrand depends only on the one-dimensional log-ratio $Z_t = X_{1,t} - X_{2,t}$. The bivariate killing problem thus reduces to a *univariate* first-exit problem for Z under $\mathbb{P}^{(k)}$, which is again a Lévy process. This reduction is purely a matter of analytical tractability: no information about the joint dynamics is discarded, since the tilted exponent $\psi_Z^{(k)}$ encodes the full marginal law of Z under $\mathbb{P}^{(k)}$.

Laplace transform and the resolvent equation. With the problem reduced to one dimension, the killed expectation $U(t,z) = \mathbb{E}_z^{(k)}[g_k(Z_t)\,\mathbf{1}_{\{\tau>t\}}]$ satisfies a PIDE in (t,z). Following Sepp (2004), applying the Laplace transform in t converts this time-evolution PIDE into a stationary *ordinary* integro-differential equation (OIDE) in z alone, with the Laplace parameter q playing the role of a discount rate and the initial condition $g_k(z)$ appearing as the forcing. The Kou structure then reduces the OIDE to an effective ODE: because exponential functions are eigenfunctions of the Kou generator, every term in the integro-differential operator acts algebraically on an exponential ansatz, and the resolvent equation becomes a finite-dimensional algebraic system. Boundary conditions at the barrier $z = -h_0$ supply three equations for the three admissible decaying modes, closing the system. Time-domain quantities are finally recovered by numerical Bromwich inversion.

The key object is the *initial log-health* $h_0 := \log(b\,w_1 S_{1,0}/w_2 S_{2,0})$, which encodes the initial distance of the position from the liquidation boundary. Under the unit-book-value normalisation $w_1 S_{1,0} - w_2 S_{2,0} = 1$, the two portfolio weights are determined entirely by h_0, reducing the optimisation to a scalar problem. The optimal $h_0^\star$ maximises a penalised conditional mean–variance criterion:

$$\Phi(h_0) = \mathbb{E}[\Pi_T \mid \tau > T] - \tfrac{\alpha}{2}\,\mathrm{Var}(\Pi_T \mid \tau > T) - \delta\, p_{\mathrm{liq}}(T, 0; h_0),$$

balancing risk-adjusted return on paths that survive against the cost of liquidation.

The paper makes the following contributions. First, we show that the portfolio optimisation reduces to a one-dimensional problem over $h_0 \in (\ln b, \infty)$, with all performance quantities expressible as explicit functions of h_0 (Section 2). Second, we derive a semi-analytical Laplace–resolvent for the killed moments $m_k(T; h_0)$, $k = 1, 2$, under the k-tilted measure (Section 5): because the payoff kink lies strictly below the barrier when $b \in (0,1)$, the forcing is a polynomial on the entire survival domain and the resolvent reduces to a 3×3 linear system with no interface conditions. Third, we derive the survival probability by the same method with $k = 0$ (Section 6).

The remainder of the paper is organised as follows. Section 2 introduces the position, health factor, payoff, and the one-dimensional reduction to a scalar optimisation over h_0. Section 3 specifies the asset dynamics under the Kou model; Section 3.1 introduces the tilted-measure machinery. Section 4 shows how a change of measure decouples the short-leg factor from the barrier problem, reducing each moment to a one-dimensional killed expectation, and states the objective in terms of those moments. Sections 5 and 6 derive the respective resolvents. Section 7 discusses numerical inversion and stability.

2 Problem Setup

We consider a leveraged long–short portfolio in which an investor holds a long position in asset S_1 and a short position in asset S_2, financed so that the net initial value of the book equals one unit of account. The

strategy is subject to a risk-management rule that forces liquidation if the long leg falls sufficiently relative to the short leg, reflecting margin requirements or stop-loss constraints common in practice. The objective is to choose the initial portfolio weights so as to optimise a performance criterion defined on the terminal payoff, conditional on surviving the liquidation threshold.

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space. Two strictly positive asset prices S_1, S_2 evolve as

$$S_{i,t} = S_{i,0} e^{X_{i,t}}, \qquad X_{i,0} = 0, \quad i = 1, 2,$$

where X_1, X_2 are log-price processes to be specified in Section 3. The investor takes a long position of size $w_1 > 0$ in S_1 and a short position of size $w_2 > 0$ in S_2.

Health ratio and liquidation. Fix a margin parameter $b \in (0, 1)$. The *health ratio* measures the current cushion of the long leg relative to the liability of the short leg:

$$H_t := \frac{b\, w_1 S_{1,t}}{w_2 S_{2,t}}, \qquad h_t := \log H_t.$$

Liquidation is triggered at the first time the long leg can no longer cover the fraction b of the short liability:

$$\tau := \inf\{t \geq 0 : \; H_t < 1\}.$$

When $H_t \geq 1$ the position is solvent; when $H_t < 1$ it is wound down. The parameter b governs how conservatively the threshold is set: a larger b triggers liquidation earlier and reduces the risk of a disorderly unwind.

Log-ratio dynamics. Since

$$h_t = \log\Big(\frac{b\, w_1 S_{1,0}}{w_2 S_{2,0}}\Big) + (X_{1,t} - X_{2,t}),$$

define the initial log-health and the log-ratio process by

$$h_0 := \log\Big(\frac{b\, w_1 S_{1,0}}{w_2 S_{2,0}}\Big), \qquad Z_t := X_{1,t} - X_{2,t},$$

so that $h_t = h_0 + Z_t$ and the liquidation time is equivalently

$$\tau = \inf\{t \geq 0 : \; Z_t < -h_0\}.$$

The one-dimensional process Z thus inherits all the relevant risk: the position survives if and only if Z stays above the level $-h_0$.

Terminal payoff. The gross profit at horizon T, contingent on survival, is

$$\Pi_T := (w_1 S_{1,T} - w_2 S_{2,T})^+ \, \mathbf{1}_{\{\tau > T\}}.$$

The positive part reflects that the investor closes the position only if it is in profit at T; the indicator $\mathbf{1}_{\{\tau>T\}}$ reflects that the position must have survived to maturity. Using the definition of H_T, one obtains the following factorization.

Lemma 2.1 (Payoff factorization)**.**

$$\Pi_T = \frac{w_2}{b}\, S_{2,0} e^{X_{2,T}} \, (e^{h_0 + Z_T} - b)^+ \, \mathbf{1}_{\{\tau > T\}}. \tag{1}$$

This separates the payoff into a numeraire factor driven by S_2, and a barrier-killed option on the log-ratio Z with moneyness controlled by h_0.

2.1 One-Dimensional Reduction and Optimisation

The two weights (w_1, w_2) represent two degrees of freedom. A normalization reduces this to a single scalar, and the structure of the payoff then identifies h_0 as the natural optimisation variable.

Normalization. We impose the unit-initial-value constraint

$$w_1 S_{1,0} - w_2 S_{2,0} = 1,$$

which fixes the net book value at inception. Setting $x := w_2 S_{2,0}$, one has $w_1 S_{1,0} = 1+x$ and $e^{h_0} = b(1+x)/x$. Solving for x,

$$x = \frac{b}{e^{h_0} - b},$$

which yields the weight map

$$w_2(h_0) = \frac{b}{S_{2,0}(e^{h_0} - b)}, \qquad w_1(h_0) = \frac{e^{h_0}}{S_{1,0}(e^{h_0} - b)}. \tag{2}$$

Positivity of both weights requires $e^{h_0} > b$, i.e.

$$h_0 > \ln b.$$

Optimisation over h_0. Since every quantity of interest — the expected payoff, its variance, and the liquidation probability — is determined by h_0 through (2) and the killed process (Z, τ), the two-weight portfolio problem reduces to a one-dimensional optimisation:

$$\max_{h_0 \in (\ln b, \infty)} \Phi(h_0)\,,$$

where Φ is a scalar performance criterion, for example the Sharpe ratio, expected return subject to a liquidation probability constraint, or a mean–variance objective. The optimal weights are then recovered by substituting $h_0^\star$ back into (2).

The specific criterion studied in this paper is a penalised conditional mean–variance objective. Let $\mathbb{E}[\,\cdot \mid \tau > T]$ denote expectation conditional on non-liquidation. The objective is

$$\Phi(h_0) := \mathbb{E}[\Pi_T \mid \tau > T] - \tfrac{\alpha}{2}\,\mathrm{Var}(\Pi_T \mid \tau > T) - \delta\, p_{\mathrm{liq}}(T, 0; h_0),$$

where $\alpha \geq 0$ is a risk-aversion coefficient and $\delta \geq 0$ penalises the probability of forced liquidation. The first two terms reward a high risk-adjusted return on the surviving paths; the third term discourages positions so aggressive that liquidation becomes likely. The interplay between the three terms means that the optimal $h_0^\star$ balances leverage (higher h_0 increases the conditional payoff but also raises liquidation risk) against downside protection.

The remainder of the paper is devoted to computing the building blocks of Φ: the killed moments $m_k(T; h_0)$ and the survival probability $p_{\mathrm{surv}}(T, 0; h_0)$, each as a closed-form function of h_0, evaluated semi-analytically via the Laplace–resolvent method developed in Section 5.

3 Asset Dynamics

We assume under $\mathbb{P}$ that each log-price satisfies the SDE

$$\mathrm{d}X_{i,t} = \mu_i^{\mathrm{eff}}\,\mathrm{d}t + \sigma_i\,\mathrm{d}B_{i,t} + \mathrm{d}\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right), \qquad i = 1, 2, \tag{3}$$

where B_1, B_2 are correlated standard Brownian motions with $\mathrm{d}\langle B_1, B_2\rangle_t = \rho\,\mathrm{d}t$, N_i is a Poisson process with intensity λ_i independent of (B_1, B_2), and the jump sizes $J_i^{(n)}$ are i.i.d. with the Kou double-exponential law. The jump density, characteristic function, and moment generating function are

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}, \qquad \varphi_{J_i}(u) = \frac{p_i}{1 - iu\eta_{i,+}} + \frac{1-p_i}{1 + iu\eta_{i,-}}, \tag{4}$$

$$M_i(s) = \frac{p_i}{1 - s\eta_{i,+}} + \frac{1-p_i}{1 + s\eta_{i,-}}, \qquad s \in \left(-\tfrac{1}{\eta_{i,-}}, \tfrac{1}{\eta_{i,+}}\right). \tag{5}$$

Here $\eta_{i,+}$ and $\eta_{i,-}$ are the *means* of the positive and negative jump sizes (not exponential rates); the corresponding rates are $1/\eta_{i,+}$ and $1/\eta_{i,-}$. The constraint $\eta_{i,+} < 1$ is required for the price-jump compensator χ_i defined below to be finite.

Drift convention and effective drift. The parameter μ_i is the *continuously compounded expected asset-price growth rate*, i.e. $\mathbb{E}_{\mathbb{P}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$. Define the price-jump compensator

$$\chi_i := M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1-\eta_{i,+}} + \frac{1-p_i}{1+\eta_{i,-}} - 1. \tag{6}$$

Since $\mathbb{E}_{\mathbb{P}}[e^{X_{i,t}}] = \exp(t[\mu_i^{\text{eff}} + \frac{1}{2}\sigma_i^2 + \lambda_i\chi_i])$, matching to $e^{\mu_i t}$ gives the log-price drift coefficient

$$\mu_i^{\text{eff}} := \mu_i - \tfrac{1}{2}\sigma_i^2 - \lambda_i\chi_i. \tag{7}$$

All analytical and numerical objects in the sequel use μ_i^{eff}.

The bivariate process (X_1, X_2) is a Lévy process with joint characteristic exponent

$$\begin{aligned}\Psi(u,v) := i(\mu_1^{\text{eff}}u + \mu_2^{\text{eff}}v) - \tfrac{1}{2}\Big(\sigma_1^2u^2 + \sigma_2^2v^2 + 2\rho\sigma_1\sigma_2uv\Big)\\ + \lambda_1\big(\varphi_{J_1}(u) - 1\big) + \lambda_2\big(\varphi_{J_2}(v) - 1\big),\end{aligned} \tag{8}$$

so that $\mathbb{E}_{\mathbb{P}}[e^{i(uX_{1,t}+vX_{2,t})}] = e^{t\Psi(u,v)}$.

Parameter summary. Table 1 collects all model parameters and their roles.

Symbol	Asset	Economic meaning / role	Constraint
μ_i	$i = 1,2$	Continuously compounded expected price growth rate: $\mathbb{E}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$	
σ_i	$i = 1,2$	Diffusion volatility of log-price X_i	$\sigma_i > 0$
ρ	—	Correlation between the two Brownian drivers	$\lvert\rho\rvert < 1$
λ_i	$i = 1,2$	Jump arrival intensity (expected jumps per unit time)	$\lambda_i > 0$
p_i	$i = 1,2$	Probability that a jump in X_i is upward	$p_i \in (0,1)$
$\eta_{i,+}$	$i = 1,2$	Mean size of upward jumps in X_i	$0 < \eta_{i,+} < 1$
$\eta_{i,-}$	$i = 1,2$	Mean size of downward jumps in X_i	$\eta_{i,-} > 0$
χ_i	$i = 1,2$	Price-jump compensator $M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1]$; enters μ_i^{eff} via (7)	
μ_i^{eff}	$i = 1,2$	Actual drift coefficient in the log-price SDE (3)	
$\Psi(u,v)$	—	Joint characteristic exponent; $e^{T\Psi(0,-ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ appears in the moment formula	

Table 1: Model parameters and derived quantities. The upper block contains freely chosen inputs; the lower block contains derived quantities used throughout the analysis.

3.1 Tilted Dynamics

For each integer $k \geq 1$, define the k-tilted measure $\mathbb{P}^{(k)}$ by

$$\left.\frac{\mathrm{d}\mathbb{P}^{(k)}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_t} = \frac{e^{kX_{2,t}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,t}}]} = \exp(kX_{2,t} - t\,\Psi(0,-ik)), \qquad k\eta_{2,+} < 1.$$

Lemma 3.1 (Lévy exponent under $\mathbb{P}^{(k)}$)**.** Under $\mathbb{P}^{(k)}$, the log-ratio $Z = X_1 - X_2$ is again a Lévy process with exponent

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)}s + \tfrac{1}{2}\sigma_Z^2s^2 + \lambda_1\big(M_1(s) - 1\big) + \lambda_2\big(M_2(k-s) - M_2(k)\big), \tag{9}$$

where

$$\sigma_Z^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \geq 0, \qquad \mu_Z^{(k)} := (\mu_1^{\text{eff}} - \mu_2^{\text{eff}}) - k(\sigma_2^2 - \rho\sigma_1\sigma_2).$$

In particular, the generator satisfies $\mathcal{L}^{(k)}e^{sz} = \psi_Z^{(k)}(s)e^{sz}$.

This eigenrelation is the key identity used in Section 5 to build the resolvent equation.

Remark 3.1 (Superscript (k) convention)**.** A superscript (k) indicates evaluation under $\mathbb{P}^{(k)}$. The untilted case $k = 0$ recovers the original dynamics of Z under $\mathbb{P}$; $k = 1$ and $k = 2$ appear in the first and second moment computations respectively.

For the Kou structure, the tilt transforms the jump rates of Z into the k-dependent phase rates

$$r_{1,+}^{(k)} := \frac{1}{\eta_{1,+}}, \quad r_{1,-}^{(k)} := \frac{1}{\eta_{1,-}}, \quad r_{2,+}^{(k)} := \frac{1}{\eta_{2,-}} + k, \quad r_{2,-}^{(k)} := \frac{1}{\eta_{2,+}} - k.$$

Because $Z = X_1 - X_2$, positive jumps of X_2 appear as downward jumps of Z: the tilt by e^{kX_2} changes the downward phase rate from $1/\eta_{2,+}$ to $1/\eta_{2,+} - k$, which is positive precisely when $k\eta_{2,+} < 1$. The associated phase operators integrate f against the exponential jump kernel:

$$K_{j,+}^{(k)}[f](z) := \int_0^\infty f(z+y)\, r_{j,+}^{(k)} e^{-r_{j,+}^{(k)} y}\, \mathrm{d}y, \qquad K_{j,-}^{(k)}[f](z) := \int_{-\infty}^0 f(z+y)\, r_{j,-}^{(k)} e^{r_{j,-}^{(k)} y}\, \mathrm{d}y,$$

for $j = 1, 2$.

Lemma 3.2 (Phase operators on exponentials)**.** For any γ with $\gamma < r_{j,+}^{(k)}$ and $\gamma > -r_{j,-}^{(k)}$,

$$K_{j,+}^{(k)}[e^{\gamma\cdot}](z) = \frac{r_{j,+}^{(k)}}{r_{j,+}^{(k)} - \gamma} e^{\gamma z}, \qquad K_{j,-}^{(k)}[e^{\gamma\cdot}](z) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma} e^{\gamma z}. \tag{10}$$

The nonlocal jump integral thus reduces to a scalar multiple of the same exponential. This is the key algebraic identity exploited in Section 5 to convert the PIDE into a finite-dimensional linear system.

Tilted parameter summary. Table 2 collects the quantities introduced in this subsection.

Symbol	Definition	Role
$\mathbb{P}^{(k)}$	$\mathrm{d}\mathbb{P}^{(k)}/\mathrm{d}\mathbb{P} \propto e^{kX_{2,t}}$	Measure under which moment reduction is carried out
k	Integer, $k \in \{0, 1, 2\}$	Tilting order; $k = 1, 2$ for moments, $k = 0$ for survival probability
$\psi_Z^{(k)}(s)$	Lévy exponent of Z under $\mathbb{P}^{(k)}$	Characteristic root equation $\psi_Z^{(k)}(\gamma) = q$ governs the resolvent
σ_Z^2	$\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$	Variance of the Brownian part of Z; appears in $\psi_Z^{(k)}$
$\mu_Z^{(k)}$	$(\mu_1^{\mathrm{eff}} - \mu_2^{\mathrm{eff}}) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$	Drift of Z under $\mathbb{P}^{(k)}$; appears in $\psi_Z^{(k)}$
$\mathcal{L}^{(k)}$	Generator of Z under $\mathbb{P}^{(k)}$	$\mathcal{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$; used in the resolvent PIDE
$r_{j,\pm}^{(k)}$	Phase rates (see above)	Exponential rates of the jump components of Z under $\mathbb{P}^{(k)}$
$K_{j,\pm}^{(k)}$	Phase operators (see above)	Convert jump integrals to scalars on exponential ansatz

Table 2: Tilted-dynamics quantities. Subscript $j = 1$ refers to the contribution from X_1 jumps; $j = 2$ to that from X_2 jumps (sign-reversed because $Z = X_1 - X_2$).

4 Moments via Measure Change

Computing $\mathbb{E}_{\mathbb{P}}[\Pi_T^k]$ directly under $\mathbb{P}$ is difficult because the payoff couples the joint process (X_1, X_2) with the barrier event $\{\tau > T\}$, producing an expectation that is neither separable nor analytically tractable. The key step is a change of measure that absorbs the $e^{kX_{2,T}}$ factor — which carries the short-leg contribution — into the density, leaving a purely one-dimensional killed expectation for the log-ratio Z under the k-tilted measure $\mathbb{P}^{(k)}$. The moment formula then factorises into the known scalar $e^{T\Psi(0,-ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ and the barrier-killed expectation $m_k(T; h_0)$ under $\mathbb{P}^{(k)}$, which is tractable via the Laplace–resolvent of Section 5.

Define

$$g_k(z; h_0) := (e^{h_0+z} - b)^k.$$

Since the survival event $\{\tau > T\}$ implies $Z_T > -h_0$, and $b \in (0,1)$ gives $z_*(h_0) := \ln b - h_0 < -h_0$, we have $e^{h_0+Z_T} > 1 > b$ on $\{\tau > T\}$. The positive part in (1) is therefore redundant on the survival set, and g_k is a polynomial in e^{Z_T}. From (1),

$$\Pi_T^k = \left(\frac{w_2}{b}\right)^k S_{2,0}^k e^{kX_{2,T}}\, g_k(Z_T; h_0)\, \mathbf{1}_{\{\tau>T\}}.$$

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{w_2}{b}\right)^k S_{2,0}^k\, e^{T\Psi(0,-ik)}\, \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T; h_0)\, \mathbf{1}_{\{\tau>T\}}\Big]. \tag{11}$$

Proposition 4.1 (First two moments)**.** Assume $k = 1, 2$ satisfy $k\eta_{2,+} < 1$. Define

$$m_k(T; h_0) := \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T; h_0)\, \mathbf{1}_{\{\tau>T\}}\Big].$$

Then

$$M_k(h_0) := \mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0} - b)^k}\, m_k(T; h_0),$$

where $m_k(T; h_0) = U^{(k)}(T, 0; h_0)$ is computed via the Laplace–resolvent method of Section 5.

Conditional moments and portfolio objective. Let $p_{\text{surv}}(h_0) := \mathbb{P}(\tau > T)$ and $p_{\text{liq}}(h_0) := 1 - p_{\text{surv}}(h_0)$. Since Π_T already contains the survival indicator $\mathbf{1}_{\{\tau>T\}}$, the conditional moments on surviving paths are

$$\mathbb{E}[\Pi_T \mid \tau > T] = \frac{M_1(h_0)}{p_{\text{surv}}(h_0)}, \qquad \text{Var}(\Pi_T \mid \tau > T) = \frac{M_2(h_0)}{p_{\text{surv}}(h_0)} - \left(\frac{M_1(h_0)}{p_{\text{surv}}(h_0)}\right)^2.$$

The mean–variance–liquidation objective is therefore

$$\Phi(h_0) = \frac{M_1(h_0)}{p_{\text{surv}}(h_0)} - \frac{\alpha}{2}\left[\frac{M_2(h_0)}{p_{\text{surv}}(h_0)} - \left(\frac{M_1(h_0)}{p_{\text{surv}}(h_0)}\right)^2\right] - \delta\, p_{\text{liq}}(h_0), \tag{12}$$

and the portfolio optimization reduces to the one-dimensional problem

$$\max_{h_0 \in (\ln b, \infty)} \Phi(h_0),$$

after which the optimal weights are recovered from (2).

Remark 4.1 (No-shorting limit)**.** As $h_0 \to \infty$ the weight map gives $w_2 S_{2,0} \to 0$ and $w_1 S_{1,0} \to 1$, so liquidation becomes impossible and the portfolio converges to a unit long position in asset 1. Hence

$$\lim_{h_0 \to \infty} \mathbb{E}[\Pi_T] = \mathbb{E}[e^{X_{1,T}}] = e^{\mu_1 T},$$

which provides a useful diagnostic: the formula for $M_1(h_0)$ must satisfy this limit exactly.

The remainder of the analysis is devoted to computing $m_1(T; h_0)$, $m_2(T; h_0)$, and $p_{\text{surv}}(h_0)$ via the Laplace–resolvent construction.

5 Laplace–Resolvent Construction

Fix $k \in \{1, 2\}$. Define the killed moment function

$$U(t, z) := \mathbb{E}_z^{(k)}\left[g_k(Z_t; h_0)\, \mathbf{1}_{\{\tau(h_0)>t\}}\right], \qquad z > -h_0,$$

with $U(t,z)=0$ for $z \le -h_0$. This satisfies the PIDE

$$\partial_t U = \mathcal{L}^{(k)} U, \qquad z > -h_0,$$

with initial condition $U(0,z) = g_k(z;h_0)$. Define the Laplace transform

$$\widehat{U}(q,z) := \int_0^\infty e^{-qt} U(t,z)\,\mathrm{d}t, \qquad q > 0.$$

Applying it to both sides of the PIDE proceeds in two steps.

Left-hand side. Integrate by parts in t:

$$\int_0^\infty e^{-qt}\,\partial_t U(t,z)\,\mathrm{d}t = \Big[e^{-qt}U(t,z)\Big]_0^\infty + q\int_0^\infty e^{-qt}U(t,z)\,\mathrm{d}t = -U(0,z) + q\widehat{U}(q,z),$$

where the boundary term at $t=\infty$ vanishes by integrability.

Right-hand side. Since $\mathcal{L}^{(k)}$ acts only in z, it commutes with the t-integral:

$$\int_0^\infty e^{-qt}\,\mathcal{L}^{(k)} U(t,z)\,\mathrm{d}t = \mathcal{L}^{(k)}\widehat{U}(q,z).$$

Equating and substituting the initial condition $U(0,z) = g_k(z;h_0)$ gives the resolvent equation

$$(q - \mathcal{L}^{(k)})\widehat{U}(q,z) = g_k(z;h_0), \qquad z > -h_0, \tag{13}$$

with $\widehat{U}(q,z) = 0$ for $z \le -h_0$.

Assumption 5.1 (Generic six-root regime)**.** Fix $k \in \{0,1,2\}$ and $q > 0$. Assume:

(i) $\sigma_Z > 0$;

(ii) the characteristic equation $\psi_Z^{(k)}(\gamma) = q$ has six simple roots $\gamma_1(q),\dots,\gamma_6(q)$, ordered so that

$$\mathrm{Re}(\gamma_{1,2,3}) > 0, \qquad \mathrm{Re}(\gamma_{4,5,6}) < 0;$$

(iii) $q \ne \psi_Z^{(k)}(j)$ for $j = 0,\dots,k$.

Forcing expansion. Since $b \in (0,1)$ and $z > -h_0$ together imply $e^{h_0+z} > 1 > b$, the payoff kernel is a polynomial in e^z:

$$g_k(z;h_0) = \sum_{j=0}^{k} c_{k,j}(h_0)\,e^{jz}, \qquad c_{k,j}(h_0) := \binom{k}{j}(-1)^{k-j} b^{k-j} e^{jh_0}.$$

Lemma 5.1 (Particular solution)**.** Under Assumption 5.1(iii),

$$\widehat{U}_{\mathrm{part}}(q,z;h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)}\,e^{jz} \tag{14}$$

is a particular solution to (13).

Proof. Substitute the forcing expansion into the right-hand side of (13):

$$(q - \mathcal{L}^{(k)})\widehat{U}_{\mathrm{part}} = \sum_{j=0}^{k} c_{k,j}(h_0)\,e^{jz}.$$

Since the sum is finite and $\mathcal{L}^{(k)}$ is linear, it suffices to find, for each j, a function v_j satisfying

$$(q - \mathcal{L}^{(k)})v_j(z) = c_{k,j}(h_0)\,e^{jz}.$$

Propose the exponential ansatz $v_j(z) = A_j\, e^{jz}$. By the eigenrelation (**??**),

$$\mathcal{L}^{(k)}[A_j e^{jz}] = A_j\, \psi_Z^{(k)}(j)\, e^{jz},$$

so

$$(q - \mathcal{L}^{(k)})[A_j e^{jz}] = A_j \big(q - \psi_Z^{(k)}(j)\big) e^{jz}.$$

Matching to $c_{k,j}(h_0)\, e^{jz}$ and using Assumption 5.1(iii) to ensure $q - \psi_Z^{(k)}(j) \neq 0$ gives

$$A_j = \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)}.$$

Summing over $j = 0, \ldots, k$ yields (14). □

Lemma 5.2 (Homogeneous solution)**.** Under Assumption 5.1(i)–(ii), the general solution to $(q - \mathcal{L}^{(k)})u = 0$ on $z > -h_0$ that is admissible at $+\infty$ is

$$\widehat{U}_{\text{hom}}(q, z; h_0) = \sum_{m=1}^{3} B_m^{(k)}(q; h_0)\, e^{\gamma_{m+3}(q)(z+h_0)}, \tag{15}$$

where $B_1^{(k)}, B_2^{(k)}, B_3^{(k)}$ are free constants.

Proof. Step 1: exponential ansatz. Seek solutions of the form $u(z) = e^{\gamma z}$. By the eigenrelation (**??**),

$$\mathcal{L}^{(k)} e^{\gamma z} = \psi_Z^{(k)}(\gamma)\, e^{\gamma z},$$

so $(q - \mathcal{L}^{(k)})e^{\gamma z} = (q - \psi_Z^{(k)}(\gamma))e^{\gamma z}$. This vanishes if and only if γ satisfies the characteristic equation

$$\psi_Z^{(k)}(\gamma) = q.$$

Step 2: root count. The exponent $\psi_Z^{(k)}$ is the sum of a quadratic term $\frac{1}{2}\sigma_Z^2\gamma^2$ (non-degenerate by Assumption 5.1(i)) and rational terms from the Kou jumps. Clearing denominators yields a degree-six polynomial in γ, so there are exactly six roots $\gamma_1, \ldots, \gamma_6$ (counted with multiplicity). Under Assumption 5.1(ii) all six are simple and split as $\mathrm{Re}(\gamma_{1,2,3}) > 0$ and $\mathrm{Re}(\gamma_{4,5,6}) < 0$.

Step 3: admissibility. On the half-line $z > -h_0$, the solution must remain bounded as $z \to +\infty$. Since $e^{\gamma z} \to \infty$ whenever $\mathrm{Re}(\gamma) > 0$, the roots $\gamma_1, \gamma_2, \gamma_3$ are inadmissible. Only $\gamma_4, \gamma_5, \gamma_6$ are retained.

Step 4: centering and general solution. Writing $e^{\gamma_{m+3} z} = e^{\gamma_{m+3}(z+h_0)} \cdot e^{-\gamma_{m+3} h_0}$ and absorbing the constant factor into the free coefficient, the general admissible homogeneous solution is

$$\widehat{U}_{\text{hom}}(q, z; h_0) = \sum_{m=1}^{3} B_m^{(k)}(q; h_0)\, e^{\gamma_{m+3}(q)(z+h_0)},$$

where $B_1^{(k)}, B_2^{(k)}, B_3^{(k)}$ are free constants to be determined by the boundary conditions. □

The general solution on $z > -h_0$ is then

$$\widehat{U}(q, z; h_0) = \widehat{U}_{\text{part}}(q, z; h_0) + \widehat{U}_{\text{hom}}(q, z; h_0).$$

Proposition 5.1 (Barrier resolvent)**.** Under Assumption 5.1, the coefficients $\mathbf{B}^{(k)}(q; h_0) := (B_1^{(k)}, B_2^{(k)}, B_3^{(k)})^\top$ solve the linear system

$$\mathbf{M}_{\text{bar}}^{(k)}(q)\, \mathbf{B}^{(k)}(q; h_0) = -\mathbf{b}^{(k)}(q; h_0),$$

with barrier matrix

$$\mathbf{M}_{\text{bar}}^{(k)}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_4(q)} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_5(q)} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_6(q)} \\ \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_4(q)} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_5(q)} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_6(q)} \end{pmatrix},$$

and forcing vector

$$\mathbf{b}^{(k)}(q;h_0) = \begin{pmatrix} \widehat{U}_{\text{part}}(q,-h_0;h_0) \\ K_{1,-}^{(k)}\left[\widehat{U}_{\text{part}}(q,\cdot\,;h_0)\right](-h_0) \\ K_{2,-}^{(k)}\left[\widehat{U}_{\text{part}}(q,\cdot\,;h_0)\right](-h_0) \end{pmatrix} = \begin{pmatrix} \displaystyle\sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)}\, e^{-jh_0} \\ \displaystyle\sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)}\, \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)}+j}\, e^{-jh_0} \\ \displaystyle\sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)}\, \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)}+j}\, e^{-jh_0} \end{pmatrix}.$$

Proof. Step 1: boundary conditions from killing. The killing condition $\widehat{U}(q,z;h_0)=0$ for $z \le -h_0$ imposes, at $z=-h_0^+$,

(i) $\widehat{U}(q,-h_0^+;h_0)=0$;

(ii) $K_{j,-}^{(k)}\left[\widehat{U}(q,\cdot\,;h_0)\right](-h_0^+)=0, \quad j=1,2.$

Condition (i) is continuity at the barrier. Conditions (ii) arise because $K_{j,-}^{(k)}[f](z)=\int_{-\infty}^{0} f(z+y)\, r_{j,-}^{(k)} e^{r_{j,-}^{(k)} y}\mathrm{d}y$ integrates f over landing points $z+y<z$; for $z=-h_0^+$ every landing point satisfies $z+y<-h_0$, where $\widehat{U}=0$, so the integral vanishes.

Step 2: substitute the full solution. Write $\widehat{U}=\widehat{U}_{\text{part}}+\widehat{U}_{\text{hom}}$ and insert into each condition. Since the conditions are linear, each splits as

$$[\text{condition on } \widehat{U}_{\text{hom}}] = -[\text{condition on } \widehat{U}_{\text{part}}].$$

Step 3: condition (i). Evaluating at $z=-h_0$ and using $e^{\gamma_{m+3}(q)\cdot 0}=1$,

$$\widehat{U}_{\text{hom}}(q,-h_0;h_0) = \sum_{m=1}^{3} B_m^{(k)} = -\,\widehat{U}_{\text{part}}(q,-h_0;h_0),$$

which is the first row of the system.

Step 4: conditions (ii). Apply $K_{j,-}^{(k)}$ to $\widehat{U}_{\text{hom}}$. By the phase-operator formula (10),

$$K_{j,-}^{(k)}\left[e^{\gamma_{m+3}(\cdot+h_0)}\right](-h_0) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)}+\gamma_{m+3}(q)}\, e^{\gamma_{m+3}(q)\cdot 0} = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)}+\gamma_{m+3}(q)},$$

giving row $j+1$ of the system:

$$\sum_{m=1}^{3} \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)}+\gamma_{m+3}(q)}\, B_m^{(k)} = -K_{j,-}^{(k)}\left[\widehat{U}_{\text{part}}(q,\cdot\,;h_0)\right](-h_0).$$

Applying (10) to $\widehat{U}_{\text{part}}(q,z;h_0)=\sum_j \frac{c_{k,j}}{q-\psi_Z^{(k)}(j)} e^{jz}$ yields the explicit entries of $\mathbf{b}^{(k)}$ stated above.

Step 5: matrix form and solution. Collecting the three conditions into a single system gives $\mathbf{M}_{\text{bar}}^{(k)}(q)\,\mathbf{B}^{(k)} = -\mathbf{b}^{(k)}$, and assuming $\mathbf{M}_{\text{bar}}^{(k)}(q)$ invertible,

$$\mathbf{B}^{(k)}(q;h_0) = -\big(\mathbf{M}_{\text{bar}}^{(k)}(q)\big)^{-1}\mathbf{b}^{(k)}(q;h_0). \qquad \square$$

Laplace-domain moment. Evaluating $\widehat{U}$ at the initial state $Z_0=0$ gives

$$\widehat{U}(q,0;h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)} + \sum_{m=1}^{3} B_m^{(k)}(q;h_0)\, e^{\gamma_{m+3}(q)h_0}. \tag{16}$$

The killed moment is then recovered by Laplace inversion:

$$m_k(T;h_0) = \mathcal{L}_{q\to T}^{-1}\left[\widehat{U}(q,0;h_0)\right].$$

6 Probability of Liquidation

The same resolvent architecture yields the probability of liquidation. Define

$$p_{\text{surv}}(t, z; h_0) := \mathbb{P}_z(\tau(h_0) > t), \qquad p_{\text{liq}}(t, z; h_0) := 1 - p_{\text{surv}}(t, z; h_0).$$

Let $\mathcal{L}^{(0)}$ denote the generator of Z under the untilted measure, with exponent $\psi_Z^{(0)}$, and define

$$\widehat{U}_{\text{surv}}(q, z; h_0) := \int_0^\infty e^{-qt} p_{\text{surv}}(t, z; h_0)\, \mathrm{d}t.$$

The problem is

$$(q - \mathcal{L}^{(0)})\widehat{U}_{\text{surv}}(q, z; h_0) = 1, \qquad z > -h_0,$$

with

$$\widehat{U}_{\text{surv}}(q, z; h_0) = 0, \qquad z \le -h_0.$$

The domain is the half-line $(-h_0, \infty)$, the boundary is absorbing at $z = -h_0$, and there is no interface because the source is constant.

Proposition 6.1 (Survival resolvent and liquidation probability)**.** Let $\gamma_4^{(0)}(q), \gamma_5^{(0)}(q), \gamma_6^{(0)}(q)$ be the three roots of

$$\psi_Z^{(0)}(\gamma) = q$$

with negative real part. Then

$$\widehat{U}_{\text{surv}}(q, z; h_0) = \frac{1}{q} + C_1(q; h_0)e^{\gamma_4^{(0)}(q)(z+h_0)} + C_2(q; h_0)e^{\gamma_5^{(0)}(q)(z+h_0)} + C_3(q; h_0)e^{\gamma_6^{(0)}(q)(z+h_0)}.$$

The coefficients are determined by

$$\widehat{U}_{\text{surv}}(q, -h_0^+; h_0) = 0, \qquad K_{1,-}^{(0)}[\widehat{U}_{\text{surv}}](-h_0^+) = 0, \qquad K_{2,-}^{(0)}[\widehat{U}_{\text{surv}}](-h_0^+) = 0.$$

Equivalently,

$$\mathbf{M}_{\text{surv}}(q)\, \mathbf{C}(q; h_0) = -\begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix},$$

where

$$\mathbf{C}(q; h_0) := \begin{pmatrix} C_1(q; h_0) \\ C_2(q; h_0) \\ C_3(q; h_0) \end{pmatrix},$$

and

$$\mathbf{M}_{\text{surv}}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_4^{(0)}(q)} & \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_5^{(0)}(q)} & \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_6^{(0)}(q)} \\ \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_4^{(0)}(q)} & \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_5^{(0)}(q)} & \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_6^{(0)}(q)} \end{pmatrix}.$$

At the initial state $Z_0 = 0$,

$$\widehat{U}_{\text{surv}}(q, 0; h_0) = \frac{1}{q} + C_1(q; h_0)e^{\gamma_4^{(0)}(q)h_0} + C_2(q; h_0)e^{\gamma_5^{(0)}(q)h_0} + C_3(q; h_0)e^{\gamma_6^{(0)}(q)h_0}.$$

Hence

$$p_{\text{surv}}(T, 0; h_0) = \mathcal{L}_{q\to T}^{-1}\left[\widehat{U}_{\text{surv}}(q, 0; h_0)\right], \qquad p_{\text{liq}}(T, 0; h_0) = 1 - p_{\text{surv}}(T, 0; h_0).$$

Proof. Step 1: resolvent equation. Taking the Laplace transform of the PIDE $\partial_t p_{\text{surv}} = \mathcal{L}^{(0)} p_{\text{surv}}$ with initial condition $p_{\text{surv}}(0, z; h_0) = 1$ and killing condition $p_{\text{surv}} = 0$ for $z \le -h_0$, exactly as in Section 5 with $g_k \equiv 1$, gives

$$(q - \mathcal{L}^{(0)})\widehat{U}_{\text{surv}}(q, z; h_0) = 1, \qquad z > -h_0.$$

Step 2: particular solution. The constant forcing $f \equiv 1$ is an eigenfunction of $\mathcal{L}^{(0)}$ with eigenvalue $\psi_Z^{(0)}(0) = 0$, so the particular solution is

$$\widehat{U}_{\text{surv,part}}(q, z) = \frac{1}{q - \psi_Z^{(0)}(0)} = \frac{1}{q}.$$

Step 3: homogeneous solution. By Lemma 5.2 applied with $k = 0$, the admissible homogeneous solution on $z > -h_0$ is

$$\widehat{U}_{\text{surv,hom}}(q, z; h_0) = C_1 e^{\gamma_4^{(0)}(q)(z+h_0)} + C_2 e^{\gamma_5^{(0)}(q)(z+h_0)} + C_3 e^{\gamma_6^{(0)}(q)(z+h_0)},$$

where $\gamma_4^{(0)}, \gamma_5^{(0)}, \gamma_6^{(0)}$ are the roots of $\psi_Z^{(0)}(\gamma) = q$ with $\text{Re}(\gamma) < 0$.

Step 4: boundary conditions. The killing condition imposes, at $z = -h_0^+$:

(i) $\widehat{U}_{\text{surv}}(q, -h_0^+; h_0) = 0$;

(ii) $K_{j,-}^{(0)}[\widehat{U}_{\text{surv}}](-h_0^+) = 0, \quad j = 1, 2,$

by the same argument as in Proposition 5.1: downward jumps from $-h_0^+$ land in the killed region.

Step 5: linear system. Substituting the full solution $\widehat{U}_{\text{surv}} = 1/q + \widehat{U}_{\text{surv,hom}}$ into each condition and applying (10) to the homogeneous terms yields

$$\mathbf{M}_{\text{surv}}(q)\,\mathbf{C}(q; h_0) = -\begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix},$$

where the right-hand side entries are $K_{j,-}^{(0)}[1/q](-h_0^+) = 1/q$ (since $K_{j,-}^{(0)}$ integrates to one against the exponential density). Assuming $\mathbf{M}_{\text{surv}}(q)$ invertible, the coefficients $\mathbf{C}$ are uniquely determined.

Step 6: inversion and liquidation probability. Evaluating at $Z_0 = 0$ and inverting via the Bromwich formula gives $p_{\text{surv}}(T, 0; h_0)$. The liquidation probability follows from the complement. □

7 Bromwich Inversion and Numerical Stability

After the previous sections, the relevant Laplace-domain objects are

$$\widehat{U}^{(k)}(q, z; h_0) \qquad \text{and} \qquad \widehat{U}_{\text{surv}}(q, z; h_0).$$

They are inverted in time via the Bromwich formula:

$$U^{(k)}(t, z; h_0) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} e^{qt} \widehat{U}^{(k)}(q, z; h_0)\, \mathrm{d}q,$$

$$p_{\text{surv}}(t, z; h_0) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} e^{qt} \widehat{U}_{\text{surv}}(q, z; h_0)\, \mathrm{d}q,$$

where $\sigma > 0$ is the Bromwich shift.

The explicit formulas above are derived in the generic regime where the characteristic equation

$$\psi_Z^{(k)}(\gamma) = q$$

has six simple roots and there is no resonance at the forcing exponents:

$$q \ne \psi_Z^{(k)}(j), \qquad j = 0, \ldots, k.$$

Repeated roots occur at values satisfying

$$\psi_Z^{(k)}(\gamma_\star) = q, \qquad (\psi_Z^{(k)})'(\gamma_\star) = 0.$$

These exceptional values are isolated in the generic regime.

A practical Bromwich shift σ should therefore be chosen sufficiently positive so that the inversion nodes

$$q = \sigma + i\omega$$

remain well inside the generic six-root regime and away from

- resonance values $q = \psi_Z^{(k)}(j)$,
- repeated-root values $q = \psi_Z^{(k)}(\gamma_\star)$ with $(\psi_Z^{(k)})'(\gamma_\star) = 0$.

For each numerical node q, one should monitor

$$\min_{j=0,\dots,k} |q - \psi_Z^{(k)}(j)|, \qquad \min_{a\neq b} |\gamma_a(q) - \gamma_b(q)|,$$

and the condition number

$$\kappa\Big(\mathbf{M}_{\text{bar}}^{(k)}(q)\Big).$$

If repeated roots occur, the homogeneous basis must be enlarged to

$$e^{\gamma z}, \quad z e^{\gamma z}, \quad \dots, \quad z^{m-1} e^{\gamma z}$$

for a root of multiplicity m. If resonance occurs at a forcing exponent j, so that

$$q = \psi_Z^{(k)}(j),$$

the term

$$\frac{c_{k,j}^{(z)}(h_0)}{q - \psi_Z^{(k)}(j)} e^{jz}$$

must be replaced by a generalized particular ansatz of the form

$$z^\nu e^{jz},$$

with the minimal power ν restoring linear independence from the homogeneous solution space.

8 Bibliography

References

Kou, S. G. A jump-diffusion model for option pricing. *Management Science*, 48(8):1086–1101, 2002.

Sepp, A. Analytical pricing of double-barrier options under a double-exponential jump diffusion process. *Applied Mathematical Finance*, 11(2):109–151, 2004.

A Generalised Beta of the Second Kind Approximation to the Payoff Distribution

The semi-analytic machinery of Sections 5–6 delivers the first four killed moments $M_k(h_0)$, $k = 1, 2, 3, 4$, and the survival probability $p_{\text{surv}}(h_0)$. This appendix shows how to fit a four-parameter *Generalised Beta of the Second Kind* (GB2) distribution to these outputs.

Structure of the conditional law. On $\{\tau > T\}$ the payoff satisfies

$$Y := \Pi_T \mid \{\tau > T\} = \frac{w_2}{b}\, S_{2,0}\, e^{X_{2,T}}\, (e^{h_0+Z_T} - b).$$

Since $Z_T > -h_0$ on $\{\tau > T\}$, we have $e^{h_0+Z_T} > 1 > b$, so the factor $(e^{h_0+Z_T} - b) > 0$. However, because $e^{X_{2,T}}$ can be arbitrarily close to zero, $\inf Y = 0$, and the support is $Y \in (0, \infty)$ with no natural positive shift. The full payoff law is

$$\Pi_T \overset{d}{=} p_{\text{liq}}\, \delta_0 + p_{\text{surv}}\, Y, \tag{17}$$

where $p_{\text{liq}} = 1 - p_{\text{surv}}$.

Conditional survival moments. Since $\Pi_T^k = Y^k\, \mathbf{1}_{\{\tau > T\}}$,

$$\mu_k^+ := \mathbb{E}[Y^k] = \frac{M_k(h_0)}{p_{\text{surv}}(h_0)}, \qquad M_k(h_0) = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0} - b)^k}\, m_k(T; h_0), \tag{18}$$

from Proposition 4.1, for $k = 1, 2, 3, 4$.

GB2 distribution. We model $Y \sim \text{GB2}(a, \beta, p, q)$ with density

$$f_Y(y) = \frac{a\, y^{ap-1}}{\beta^{ap}\, B(p,q)\, \big(1 + (y/\beta)^a\big)^{p+q}}, \qquad y > 0, \tag{19}$$

where $a > 0$ is a tail/shape parameter, $\beta > 0$ is a scale parameter, $p > 0$ and $q > 0$ are left- and right-shape parameters, and $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ is the beta function.

GB2 moments. For $k < aq$, the k-th raw moment is

$$\mathbb{E}[Y^k] = \beta^k\, \frac{B(p + k/a,\ q - k/a)}{B(p,q)} = \beta^k\, \frac{\Gamma(p + k/a)\, \Gamma(q - k/a)}{\Gamma(p)\, \Gamma(q)}. \tag{20}$$

The first four moments exist if and only if $q > 4/a$.

Calibration. We fit (a, β, p, q) by matching the four conditional moments $\mu_1^+, \ldots, \mu_4^+$.

Step 1: shape parameters from scale-free ratios. Define

$$R_k := \frac{\mu_k^+}{(\mu_1^+)^k} = \frac{B(p + k/a,\ q - k/a)/B(p,q)}{\big[B(p + 1/a,\ q - 1/a)/B(p,q)\big]^k}, \qquad k = 2, 3, 4. \tag{21}$$

The ratios R_k depend only on (a, p, q), not on β. Solve

$$R_k(a, p, q) = \hat{R}_k, \qquad k = 2, 3, 4, \tag{22}$$

where $\hat{R}_k = \mu_k^+/(\mu_1^+)^k$ are the targets, subject to $a > 0$, $p > 0$, $q > 4/a$. For numerical stability, evaluate (21) in log-space:

$$\log \mathbb{E}[Y^k] = k \log \beta + \log \Gamma\big(p + \tfrac{k}{a}\big) + \log \Gamma\big(q - \tfrac{k}{a}\big) - \log \Gamma(p) - \log \Gamma(q). \tag{23}$$

Step 2: scale from the first moment. Given (a, p, q) from Step 1, recover

$$\beta = \mu_1^+\, \frac{B(p,q)}{B(p + 1/a,\ q - 1/a)}. \tag{24}$$

Full distribution and quantiles. The CDF of Π_T is

$$F_\Pi(y) = \begin{cases} 0 & y < 0, \\ p_{\text{liq}} & y = 0, \\ p_{\text{liq}} + p_{\text{surv}}\, F_Y(y) & y > 0, \end{cases}$$

where F_Y is the GB2 CDF. For a confidence level $u \in (0,1)$, the Value-at-Risk is

$$\text{VaR}(u) = \begin{cases} 0 & u \le p_{\text{liq}}, \\ F_Y^{-1}\left(\dfrac{u - p_{\text{liq}}}{p_{\text{surv}}}\right) & u > p_{\text{liq}}, \end{cases}$$

where F_Y^{-1} is the GB2 quantile function.

Consistency check. After fitting, verify that

$$\beta^k \, \frac{\Gamma(p + k/a)\, \Gamma(q - k/a)}{\Gamma(p)\, \Gamma(q)} = \mu_k^+, \qquad k = 1, 2, 3, 4.$$

Rationale for GB2. The GB2 family nests several common distributions (Burr XII, log-logistic, Pareto, Weibull, log-normal in a limit) and has four free parameters, enabling the simultaneous matching of four conditional moments. It accommodates the power-law tails that arise in Kou jump-diffusion models, which the Generalized Gamma family — having thinner stretched-exponential tails — cannot capture. The condition $q > 4/a$ provides an explicit testable constraint on whether the fourth moment of the conditional payoff is finite.